Han Lin

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Research Interests

Computer vision; controllable and interactive world modeling; image/video generation; multimodal and representation learning; planning, verification, and embodied intelligence; reinforcement learning; theory-grounded algorithms for efficient Transformers.

Education

University of North Carolina at Chapel Hill Aug. 2023 - Exp. May 2028
Ph.D. in Computer Science GPA: 4.0/4.0

- MURGe-Lab. Advised by Prof. Mohit Bansal

Columbia University 2021 - 2023
M.S. in Computer Science GPA: 4.13/4.33

- DVMM Lab. Advised by Prof. Shih-Fu Chang
- ROAM Lab. Advised by Prof. Matei Ciocarlie and Prof. Shuran Song

Relevant Courses: Learning Theory, Algorithms, Machine Learning, Unsupervised Learning, Bandits & Reinforcement Learning, Causal Inference, Computer Vision, Robotics Learning

Columbia University 2018 - 2020
M.S. in Financial Engineering GPA: 4.04/4.33

- Advised by Prof. Krzysztof Choromanski from Google DeepMind

Relevant Courses: Optimization, Combinatorial Optimization, Stochastic Models, Stochastic Calculus, Monte Carlo Methods, Statistical Inference, Bayesian Statistics, Graphical Models

Central University of Finance and Economics 2014 - 2018
B.S. in Financial Engineering (Honor Class) GPA: 90.43/100
Relevant Courses: Linear Algebra, Mathematical Analysis, Probability, Statistics, Real Analysis, Numerical Methods, Stochastic Process, Differential Equations

Preprints

1. **From Failures to Learnable Worlds: Policy-Conditioned Environment Generation for Navigation Agents**
   Yue Zhang, **Han Lin**, Zun Wang, Yidong Huang, Zaid Khan, Mohit Bansal
   arXiv Preprint 2026 [arXiv | project page | code]
2. **Seeing Isn't Knowing: Do VLMs Know When Not to Answer Spatial Questions (and Why)?**
   Yue Zhang, Zun Wang, **Han Lin**, Yonatan Bitton, Idan Szpektor, Mohit Bansal
   arXiv Preprint 2026 [arXiv | project page | code]
3. **Planning with Sketch-Guided Verification for Physics-Aware Video Generation**
   Yidong Huang, Zun Wang, **Han Lin**, Dong-Ki Kim, Shayegan Omidshafiei, Jaehong Yoon, Yue Zhang, Mohit Bansal
   arXiv Preprint 2025 [arXiv | project page | code]
4. **Training-Free Guidance in Text-to-Video Generation via Multimodal Planning and Structured Noise Initialization**
   Jialu Li*, Shoubin Yu*, **Han Lin***, Jaemin Cho, Jaehong Yoon, Mohit Bansal
   arXiv Preprint 2025 [arXiv | project page | code]

5. **Graph Kernel Attention Transformers**
   Krzysztof Choromanski*, **Han Lin***, Haoxian Chen*, Jack Parker-Holder
   arXiv Preprint 2021 [paper | code]

Publications

1. **Exploring MLLM-Diffusion Information Transfer with MetaCanvas**
   Han Lin, Xichen Pan, Ziqi Huang, Ji Hou, Jialiang Wang, Weifeng Chen, Zecheng He, Felix Juefei-Xu, Junzhe Sun, Zhipeng Fan, Ali Thabet, Mohit Bansal, Chu Wang
   NeurIPS 2026 [arXiv | project page]
2. **PhyMotion: Structured 3D Motion Reward for Physics-Grounded Human Video Generation**
   Yidong Huang*, Zun Wang*, **Han Lin**, Dong-Ki Kim, Shayegan Omidshafiei, Jaehong Yoon, Jaemin Cho, Yue Zhang, Mohit Bansal
   NeurIPS 2026 [arXiv | project page]
3. **AnchorWeave: World-Consistent Video Generation with Retrieved Local Spatial Memories**
   Zun Wang, **Han Lin**, Jaehong Yoon, Jaemin Cho, Yue Zhang, Mohit Bansal
   ECCV 2026 [arXiv | project page | code]
4. **V-Co: A Closer Look at Visual Representation Alignment via Co-Denoising**
   Han Lin, Xichen Pan, Zun Wang, Yue Zhang, Chu Wang, Jaemin Cho, Mohit Bansal
   ECCV 2026 [arXiv | code]
5. **Error-Driven Scene Editing for 3D Grounding in Large Language Models**
   Yue Zhang, Zun Wang, **Han Lin**, Jialu Li, Jianing Yang, Yonatan Bitton, Idan Szpektor, Mohit Bansal
   ECCV 2026 [arXiv | code]
6. **EPiC: Efficient Video Camera Control Learning with Precise Anchor-Video Guidance**
   Zun Wang, Jaemin Cho, Jialu Li, **Han Lin**, Jaehong Yoon, Yue Zhang, Mohit Bansal
   ICML 2026 [arXiv | project page | code]
7. **A Benchmark of Expert-Level Academic Questions to Assess AI Capabilities**
   Center for AI Safety, Scale AI & HLE Contributors Consortium
   Nature 2026 [paper | project page]
8. **Bifrost-1: Bridging Multimodal LLMs and Diffusion Models with Patch-level CLIP Latents**
   Han Lin, Jaemin Cho, Amir Zadeh, Chuan Li, Mohit Bansal
   NeurIPS 2025 [arXiv | project page | code]
9. **VEDiT: Latent Prediction Architecture for Procedural Video Representation Learning**
   Han Lin, Tushar Nagarajan, Nicolas Ballas, Mido Assran, Mojtaba Komeili, Mohit Bansal, Koustuv Sinha
   ICLR 2025 [arXiv]
10. **CTRL-Adapter: An Efficient and Versatile Framework for Adapting Diverse Controls to Any Diffusion Model**
    Han Lin*, Jaemin Cho*, Abhay Zala, Mohit Bansal
    ICLR 2025 (**Oral**, Top 1.82%) [arXiv | project page | code]
11. **DreamRunner: Fine-Grained Storytelling Video Generation with Retrieval-Augmented Motion Adaptation**
    Zun Wang, Jialu Li, **Han Lin**, Jaehong Yoon, Mohit Bansal
    AAAI 2025 [arXiv | project page | code]
12. **Fast Tree-Field Integrators: From Low Displacement Rank to Topological Transformers**
    Krzysztof Choromanski*, Arijit Sehanobish*, Somnath Basu Roy Chowdhury*, **Han Lin***, Avinava Dubey, Tamas Sarlos, Snigdha Chaturvedi
    NeurIPS 2024 [arXiv]
13. **VideoDirectorGPT: Consistent Multi-Scene Video Generation via LLM-Guided Planning**
    Han Lin, Abhay Zala, Jaemin Cho, Mohit Bansal
    COLM 2024 [arXiv | project page | code]

14. **EnvGen: Generating and Adapting Environments via LLMs for Training Embodied Agents**
Abhay Zala*, Jaemin Cho*, **Han Lin**, Jaehong Yoon, Mohit Bansal
COLM 2024 [arXiv | project page | code]

15. **DiagrammerGPT: Generating Open-Domain, Open-Platform Diagrams via LLM Planning**
Abhay Zala, **Han Lin**, Jaemin Cho, Mohit Bansal
COLM 2024 [arXiv | project page | code]

16. **Efficient Graph Field Integrators Meet Point Clouds**
Krzysztof Choromanski*, Arijit Sehanobish*, **Han Lin***, Yunfan Zhao*, Eli Berger, Alvin Pan, Tetiana Parshakova, Tianyi Zhang, David Watkins, Valerii Likhosherstov, Somnath Basu Roy Chowdhury, Avinava Dubey, Deepali Jain, Tamas Sarlos, Snigdha Chaturvedi, Adrian Weller
ICML 2023 [arXiv | code]

17. **Supervised Masked Knowledge Distillation for Few-shot Transformers**
Han Lin*, Guangxing Han*, Jiawei Ma, Shiyuan Huang, Xudong Lin, Shih-Fu Chang
CVPR 2023 [arXiv | code]

18. **Active Tactile Exploration for 3D Object Recognition**
Jingxi Xu*, **Han Lin***, Shuran Song, Matei Ciocarlie
ICRA 2023 [arXiv | project page]

19. **From Block-Toeplitz Matrices to Differential Equations on Graphs: Towards a General Theory for Scalable Masked Transformers**
Krzysztof Choromanski*, **Han Lin***, Haoxian Chen*, Tianyi Zhang, Arijit Sehanobish, Valerii Likhosherstov, Jack Parker-Holder, Tamas Sarlos, Adrian Weller, Thomas Weingarten
ICML 2022 [arXiv | code]

20. **Hybrid Random Features**
Krzysztof Choromanski*, **Han Lin***, Haoxian Chen*, Yuanzhe Ma*, Arijit Sehanobish*, Deepali Jain, Michael S Ryoo, Jake Varley, Andy Zeng, Valerii Likhosherstov, Dmitry Kalashnikov, Vikas Sindhwani, Adrian Weller
ICLR 2022 [arXiv | code]

21. **Demystifying Orthogonal Monte Carlo and Beyond**
Han Lin*, Haoxian Chen*, Tianyi Zhang, Clement Laroche, Krzysztof Choromanski
NeurIPS 2020 [arXiv | code]

* Co-First Authors, Equal Contribution

Research Experience

Meta FAIR Lab (DReaM Team) May 2026 - Exp. Dec. 2026
Research Scientist Intern
Advised by: Marjan Ghazvininejad, Xiaochuang Han, Mengchen Liu, Luke Zettlemoyer

- Self-improvement of prompt rewriter and video generation model
- On-policy distillation (OPD) for video generation

Meta MSL (Movie Gen Team) May 2025 - Dec. 2025
Research Scientist Intern
Advised by: Chu Wang, Ji Hou, Jialiang Wang, Weifeng Chen, Zecheng He, Felix Juefei-Xu, Junzhe Sun, Zhipeng Fan, Ali Thabet

- Exploring MLLM-Diffusion Information Transfer with MetaCanvas (NeurIPS 2026)

Meta FAIR Lab (JEPA Team) May 2024 - Dec. 2024
Research Scientist Intern
Advised by: Koustuv Sinha, Tushar Nagarajan, Nicolas Ballas, Mido Assran, Mojtaba Komeili

- VEDiT: Latent prediction architecture for procedural video representation learning (ICLR 2025)

Google Research Collaboration on Efficient Graph Networks 2022 - 2024
Advised by: Krzysztof Choromanski, Jack Parker-Holder, Somnath Basu Roy Chowdhury, Avinava Dubey, Tamás Sarlós

- Fast Tree-Field Integrators: From Low Displacement Rank to Topological Transformers (NeurIPS 2024)
- Efficient Graph Field Integrators Meet Point Clouds (ICML 2023)
- From Block-Toeplitz Matrices to Differential Equations on Graphs: Towards a General Theory for Scalable Masked Transformers (ICML 2022)

Google Research Collaboration on Random Feature Algorithms 2019 - 2022
Advised by: Krzysztof Choromanski, Deepali Jain, Michael S. Ryoo, Jake Varley, Andy Zeng, Dmitry Kalashnikov, Vikas Sindhwani

- Hybrid Random Features (ICLR 2022)
- Demystifying Orthogonal Monte Carlo and Beyond (NeurIPS 2020)

MURGe-Lab, UNC 2023 - Present
Research Assistant, Advised by Prof. Mohit Bansal

- Text-to-video generation, multimodal learning, and LLMs

DVMM Lab, Columbia University 2022 - 2023
Research Assistant, Advised by Prof. Shih-Fu Chang

- Supervised masked knowledge distillation for few-shot Transformers (CVPR 2023)

ROAM Lab, Columbia University 2022 - 2023
Research Assistant, Advised by Prof. Matei Ciocarlie and Prof. Shuran Song

- Active tactile exploration for 3D object recognition (ICRA 2023)

Cornell, Maryland, Max Planck Pre-doctoral Research School 2022

Industry Experience

China Merchant Securities Aug. 2020 - Aug. 2021
Option Market Making Quant Trader, Full Time

- Commodity options and futures trading and daily P&L attribution

Teaching Experience

COMS 4231 Analysis of Algorithms	Fall 2022
COMS 4732 Computer Vision II: Learning	Spring 2022
COMS 4721 Machine Learning for Data Science	Spring 2022
QMSS 5073 Machine Learning for Social Science	Fall 2021
IEOR 4007 Optimization Models & Methods for FE	Fall 2019
IEOR 4418 Transportation Analytics & Logistics	Spring 2019

Awards

CVPR 2023 Travel Award — Jun 2023
CVPR community

Honorable Mention in Mathematical Contest in Modeling — Jan 2018
The Consortium for Mathematics and Its Applications (COMAP)

Scholarship for Excellence in Organization and Leadership — Apr 2017
Central University of Finance and Economics

2nd Prize for Summer & Winter Social Practice Scholarship — Oct 2015
Central University of Finance and Economics

Technical Skills

Python, C/C++, MATLAB, R, MySQL, LaTeX
PyTorch, TensorFlow, Large-Scale TPU/GPU Training

Professional Service

Workshop Organizer: Any-To-Any Multimodal Learning Workshop, CVPR, 2026
Conference Reviewer: ICML 2022-2026, NeurIPS 2022-2026, ICLR 2024-2025, CVPR 2025-2026, ICCV 2025, ECCV 2026, WACV 2026
Conference Volunteer: RSS 2022

References

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Department of Computer Science, UNC Chapel Hill
Relationship: PhD advisor, director of UNC MURGe-Lab.

Shih-Fu Chang
(sc250@columbia.edu)
Columbia University
Relationship: master's research advisor, dean of Columbia Engineering.

Shuran Song
(shuran@stanford.edu)
Stanford University
Relationship: master's research advisor, director of REAL Lab at Stanford University.

Chu Wang
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Meta
Relationship: internship advisor, research scientist at Meta MSL.

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Google DeepMind Robotics & Columbia University
Relationship: master's advisor and research collaborator.

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Relationship: master's research advisor, director of ROAM Lab at Columbia University.

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